

1 Definitions

1.1 Supplier Module

Supplier Represents a supplier in the system. A supplier contains a registration number, ID, name, bank account, and payment terms. It also maps names to contact information, delivery day schedules, and a supplier agreement.

SupplierAgreement Represents an agreement with a supplier. The agreement maps item catalog numbers to prices. It also contains a map from item catalog numbers to their discount rules.

ContactInfo Represents a supplier contact person. Contact information contains a name, email address, and phone number.

DiscountRule Represents a discount that a supplier offers to the purchaser. A discount rule contains a name, a minimum threshold, and a discount percentage.

PaymentTerms Represents the supplier payment terms. Payment terms contain a textual description.

DeliveryDaySchedule Represents the list of days on which orders can be posted to the supplier.

SupplierOrder Represents an order from a supplier. A supplier order contains a supplier ID, order date, order ID, status, and a list of supplier items.

SupplierItem Represents an item type in a supplier order. A supplier item contains the effective price after applying a discount rule, the amount ordered, and the item catalog number.

BuyOrder Represents an automatic buy order that creates orders to a supplier with fixed items and amounts on fixed days. A buy order contains a list of supplier items, a supplier ID, and a delivery day schedule.

1.2 Inventory Module

ProductDL Represents a product in the supermarket. A product contains a name, catalog number, warehouse, shelf location, main category ID, sub-category ID, optional sub-sub-category ID, manufacturer, amount on shelf, amount in stock, list of promotions, consumer price, supply price, supplier discount, and minimum amount alert.

Warehouse Represents a warehouse in the supermarket. A warehouse contains a textual value.

ShelfLocation Represents a shelf location in the supermarket. A shelf location contains a textual value.

CategoryDL Represents an item category. A category contains a name, ID, list of sub-categories, list of promotions, and type. The category type can be main, sub, or sub-sub.

FaultyProductDL Represents a faulty product report. A faulty product report contains a report ID, product name, catalog number, shelf location, textual description, and report date.

Notification Represents a low-stock alert sent by the system when the total product amount on shelves and in stock is lower than the product minimum amount alert. A notification contains the product name, catalog number, shelf location, minimum threshold, amount in stock, and amount on shelves. It also computes the number of items required to restock.

Module	Type	Description	Priority	Risk
Inventory	Functional	The system must notify early on products that are about to run out, based on a pre-set minimal amount.	MH	Low
Inventory	Functional	Shortage notifications must include the product catalog number, location, and amount to restock.	MH	Low
Inventory	Functional	The system must allow adding products, allowing workers to track products in stock.	MH	High
Inventory	Functional	The system must store data on products: catalog number, name, category, sub-categories, location in the supermarket, manufacturer, and current amount on shelves and in storage.	MH	Low
Inventory	Functional	The system must allow viewing, updating, and deleting products from the catalog, including updating the product fields described in the product data requirements.	MH	Low
Inventory	Functional	The system must allow storing price data: catalog number, supply price, customer price, supplier discount, and customer discount.	MH	High
Inventory	Non-Functional	The system must store product data by categories. Each product will have a single category and sub-category, and may optionally have a sub-sub-category. Categories must match the product and not contradict each other.	MH	High
Inventory	Functional	The system must allow setting promotions on individual products or categories, including a date range and a discount percentage.	MH	Low
Inventory	Functional	The system should allow creating inventory reports for any number of categories. Reports include: product name, catalog number, and amount by location, both in stock and on shelves.	NTH	Low
Inventory	Functional	The system should allow reporting a faulty product, including providing a textual description of the flaw, and must store the report.	NTH	Low
Inventory	Functional	The system should allow creating a faulty items report, showing all reports within a date range.	NTH	Low
Supplier	Functional	The system must allow adding, editing, viewing, and deleting suppliers.	MH	High
Supplier	Functional	The supplier record must store the registration number, bank account, payment terms, and contact details.	MH	Low
Supplier	Functional	Each supplier can hold any number of contacts. Each contact must include a valid name, email address, and phone number. The system must allow adding, updating contact details, and removing contacts.	MH	Low
Supplier	Functional	Each supplier has an agreement that maps item catalog numbers to their agreed price.	MH	High
Supplier	Functional	The system must allow adding an item and price to the agreement, removing an item from the agreement, and updating its price.	MH	Low
Supplier	Non-Functional	Suppliers can be set with fixed delivery days, and can only receive orders on those days.	MH	Low
Supplier	Functional	Orders must be created from agreements, given quantities for each item catalog number. Items in an order must be included in the agreement.	MH	High
Supplier	Functional	All orders must be stored and able to be reviewed.	MH	Low
Supplier	Functional	Changes to agreements must affect future orders only.	MH	Low
Supplier	Functional	The system should allow posting urgent orders to suppliers with fixed delivery days, even outside their regular order days.	NTH	Low

Module	Type	Description	Priority	Risk
Supplier	Functional	The same inventory item can be supplied by multiple different suppliers.	NTH	Low
Supplier	Functional	The system should allow suppliers to define discount rules for items in the agreement, based on the purchased amount. Each rule must include a minimum threshold and a discount percentage.	NTH	Low
Both	Functional	The system must allow placing a buy order that creates automatic orders for the selected items, using preset suppliers, item catalog numbers, and quantities. Prices will be taken from the supplier agreement.	MH	High
Both	Non-Functional	If the supplier has fixed delivery days, an automatic order can only be placed on one of those days.	MH	Low
Both	Functional	The system must allow editing buy orders, including adding items and amounts, removing items, changing the fixed order date, and removing buy orders.	MH	Low
Both	Functional	Upon receiving shortage notifications, the system must find the best supplier that has an agreement for the item and create an automatic order with the amount specified in the notification.	MH	High

1.3 Changes From Prior Assignment

The following changes were made compared to the prior assignment:

- Added a definitions page that defines the main entities used by the Inventory and Supplier modules.
- Added new entities, including Notification, BuyOrder, ShelfLocation, Warehouse, and SupplierOrder.
- Added requirements regarding supplier behavior on fixed delivery days, including the restriction that suppliers cannot receive orders outside their fixed delivery days unless the order is urgent.
- Added SupplierOrder requirements and made SupplierAgreement more specific.
- Added notification behavior to the requirements.
- Clarified the Inventory module requirement regarding the category hierarchy.
- Added a promotions entity requirement for the Inventory module.
- Added requirements for automatic orders based on notifications and buy orders.

The following changes were made compared to the prior assignment. These changes were made to satisfy the new requirements, as well as to meet the expectations from Assignment 1 according to the feedback given.